

DeMorgan's theorem states that _____

Select one:

☒ a. $(AB)' = A' + B'$

☐ b. $A' + B' = A'B'$

☐ c. $(AB)' = A' + B$

☐ d. $(A + B)' = A' * B$

What logic gate gives an output of logic one if there is an odd number of 1's at the input?

Select one:

☐ XNOR gate

☒ XOR gate

☐ NAND gate

☐ NOR gate

The logical value (1) is 1 volt

Select one:

☒ a. false

☐ b. true

How many ICs are required to realize $Y = CD + EF + G$?

Select one:

☐ 5

☐ 2

☐ 4

☒ 3

DIGITAL LOGIC DESIGN LAB-5

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Which of the following Boolean expressions are false?

Select one:

- ☐ $(X+Y)(X+Z)=X+YZ$
- ☐ $XY+X'=X+Y$
- ☒ $(X+Y)(X'+Z)=X'Y+XZ$
- ☐ $X.1 = X$

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Logic gate whose output is HIGH when one or all of its inputs is LOW.

Select one:

- ☐ AND gate
- ☐ NOR gate
- ☒ NAND gate
- ☐ OR gate

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DIGITAL LOGIC DESIGN LAB-10

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9

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How many INVERTERS gates are contained in a 7404 IC?

Select one:

- ☐ 2
- ☐ 4
- ☐ 8
- ☒ 6

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Simplify

$$(A + B)(A'.B')$$

Select one:

☐ a. AB'

☒ b. 0

☐ c. AB

☐ d. 1

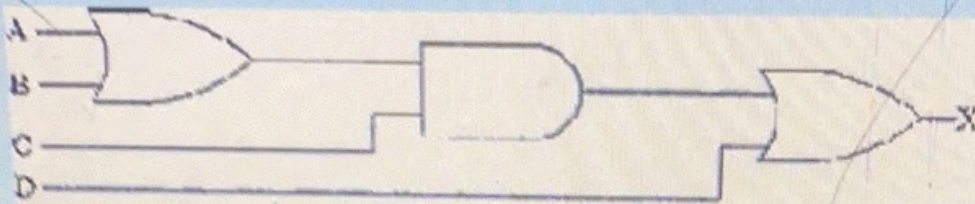
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Solve the circuit for X.



Select one:

☐

$$A + BC + D$$

☐

$$D(A + B + C)$$

☐

$$((A + B)C) + D$$

☐

$$(AC + BC)D$$

Exclusive-OR (XOR) logic gates can be constructed from what other logic gates?

Select one:

- ☐ a. OR gates and NOT gates
- ☒ b. AND gates, OR gates, and NOT gates
- ☐ c. AND gates and NOT gates
- ☐ d. OR gates only

How many NAND gates are contained in a 7400 NAND IC?

Select one:

☒ 4

☐ 8

☐ 1

☐ 2



The Boolean function $x'y' + xy + x'y$ is equivalent to

Select one:

☒ a.

$x' + y$

☐ b.

$x' + y'$

☐ c.

$x + y$

☐ d.

$x + y'$



The NOR gate output will be high if the two inputs are _____

Select one:

☒ a. 00

☐ b. 11

☐ c. 10

☐ d. 01

Which one of the following expressions does NOT represent exclusive NOR of x and y ?

Select one:



$$x \oplus y$$



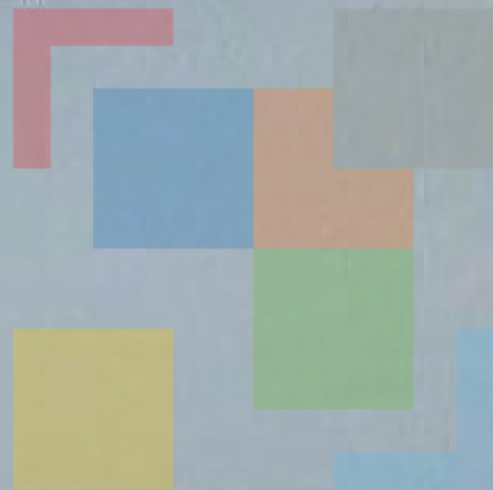
$$x \oplus y'$$



$$xy + x'y'$$



$$x \oplus y$$

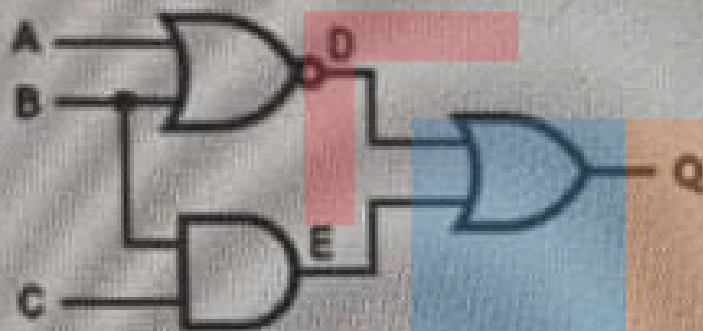


A circuit that converts the input logic level to its complement.

Select one:

- ☐ a. NAND gate with all inputs tied
- ☐ b. Inverter
- ☐ c. NOR gate with all inputs tied
- ☒ d. All mentioned

Solve the circuit for Q



Select one:



$$A'B' + BC$$



$$(A+B)' + (B'+C)'$$



$$(A'+B).(B+C)'$$



$$(A+B) + B.C$$

Exclusive-OR (XOR) logic gates can be constructed from what other logic gates?

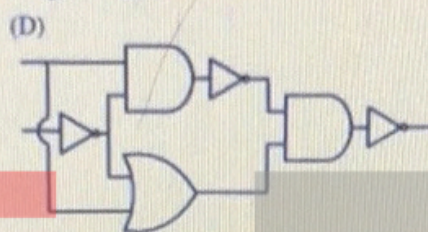
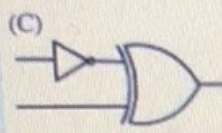
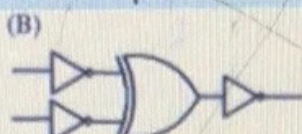
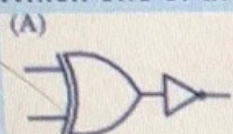
Select one:

- ☐ a. OR gates and NOT gates
- ☐ b. AND gates and NOT gates
- ☐ c. OR gates only
- ☒ d. AND gates, OR gates, and NOT gates

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Which one of the following circuits is NOT equivalent to a 2-input XNOR (exclusive NOR) gate?



Select one:

☐ B

☐ D

☒ C

☐ A

F5

F6

F7

F8

F9

F10

F11

F12

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A circuit that converts the input logic level to its complement.

Select one:

- ☒ a. NOR gate with all inputs tied
- ☐ b. All mentioned
- ☐ c. Inverter
- ☐ d. NAND gate with all inputs tied

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13-DIGITAL LOGIC DESIGN LAB-5 ▶ General ▶ Midterm Exam

The truth table represents the Boolean function

X	Y	$f(X, Y)$
0	0	0
0	1	0
1	0	1
1	1	1

Select one:

- ☐ $X+Y$
- ☐ $X \oplus Y$
- ☐ Y

☒ X

DELL

DIGITAL LOGIC DESIGN LAB-5

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A circuit that converts the input logic level to its complement.

Select one:

- ☐ a. NOR gate with all inputs tied
- ☒ b. All mentioned
- ☐ c. Inverter
- ☐ d. NAND gate with all inputs tied

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Which of the following is a form of De Morgan's theorem?

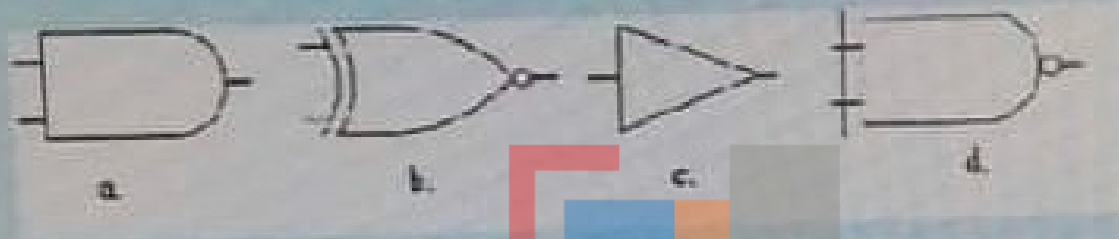
Select one:

- ☐ a. $A + B = (AB)'$
- ☐ b. $A \cdot B = A' \cdot B'$
- ☐ c. $(A + B)' = A' \cdot B'$
- ☒ d. $AB = (A + B)'$

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Which of the figures shown represents the **exclusive-NOR gate**?



Select one:

☐ A

☐ C

☒ B

☐ D

LAB-10

[Home](#) → [Second Semester 2021/2020](#)[2021/2020-2-30102213-DIGITAL LOGIC DESIGN LAB-10](#) →[General](#) → [MID QUIZ](#)

Question 3

Not yet answered

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 Flag question

According to Demorgan's Theory , which of the following is correct

Select one:

- ☐ a. $(AB)' = A' + B'$
- ☐ b. $(XYZ)' = X' + Y' + Z'$
- ☐ c. $(A + B + C)' = A'B'C'$
- ☒ d. all are correct

[Next page](#)

QUIZ NAVIGATION

1	2	3	4	5	6	7	8	9
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2213-DIGITAL LOGIC DESIGN LAB-5 ► General ► Midterm

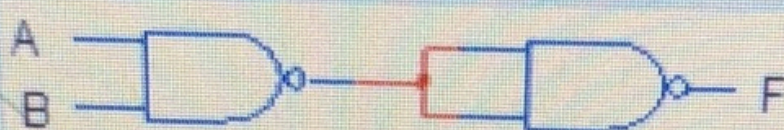
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F can be implemented using



Select one:

☐ XOR

☒ AND

☐ NOR

☐ OR

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Question 15

Not yet answered

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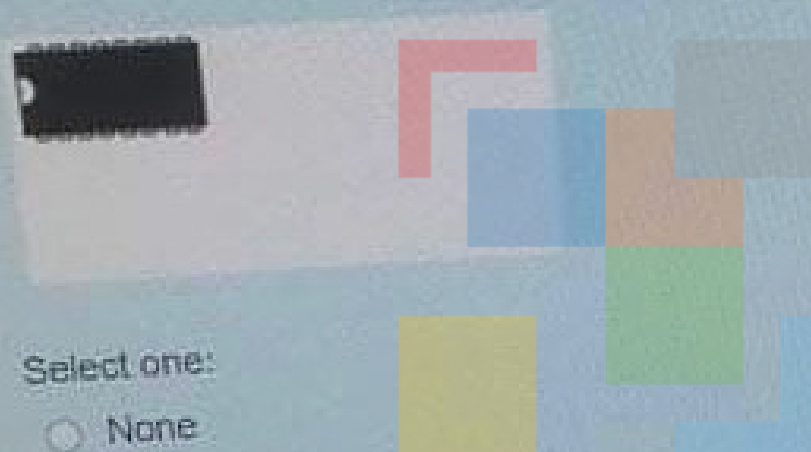
A logic gate whose output is logic zero every time one of its inputs goes to logic one.

Select one:

- ☐ XOR gate
- ☐ A and C
- ☒ NOR gate
- ☐ NAND gate

Finish att

The correct leg numbering for the following IC is ... starting from the mark on it.



Select one:

- ☐ None
- ☐ Clockwise
- ☒ Counter Clockwise
- ☐ Counter Clockwise and Clockwise are the same

3

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question

How many two-input AND and OR gates are required to implement $Y = CD + EF + G$?

Select one:

☒ a. 2, 2

☐ b. 3, 3

☐ c. 3, 2

☐ d. 2, 3